



A BPM-based Emerald Factory Information System: Redesigning Emerald Crafting Process in Myanmar

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ABSTRACT

In the emerald factory business, the crafting process plays a crucial role in driving the quality of products. However, such a process has limitations and difficulties in monitoring the working progress, determining the factory hours to complete the products, reporting the completed tasks, and reporting the defects. This causes cost and time inefficiency in managing this business. To reach this aim, this study, therefore, addresses such issues by employing business process management strategy to help analyze and redesign the existing crafting process for better performance, where the information system was developed to help improve the certain tasks. Hence, the swimlane diagram was used to visualize the business tasks and flows; moreover, the information system was developed and implemented for employees and managers accordingly. The overall structure and screenshots of the proposed system were presented in the paper. After being deployed on the cloud server, the system was used for weeks and evaluated by the users. The results show that both group of users highly accepted the system, while the managers had a significantly better attitude in analyzing and managing the emerald crafting process. Finally, the findings of this study shed light of the implications of understanding and redesigning the business process with the benefits of the information system to enhance business success.

CCS Concepts

- Information systems → Information systems applications
- Applied computing → Business process management.

Keywords

Business information system; business process redesign; emerald factory; crafting process; decision support system.

1. INTRODUCTION

The jade business offers deliverables in two different types, raw materials and finished products. Raw materials are highly demanded in China market because fine emeralds can barely be found in any place rather than Myanmar, plus Chinese technologies are far way better at crafting emeralds in different shapes.

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Over half of the business incomes becomes from selling finished products to local people and foreigners. These products are required to go through emerald crafting process (ECP) to get very fine jades. Every process in the ECP is to be carefully focused on because any small mistake can lead to big profit loss and the broken jade from the crafting becomes much less valuable suddenly. Generally, the crafting process has to deal with the issues of different material sizes and shapes which then results in no specific time the process has to be finished. Similarly, pieces of jades from breaking down a rock have many different shapes and thus setting a fix deadline to finish the crafting process seem to be impossible, especially when it comes to creating very unique shapes. Also, the broken jades from the crafting process result in big losses.

Since there are millions of unique shapes that take time to finish, a software that helps designing the 3D images was created and this software is compatible with a crafting machine. However, the software does not solve the problems of the rest of time taking crafting processes and the fact that all the broken jades are left behind without the exact profit loss data still persists. Many broken jades and fragments can be found during the crafting processes. The company lacks the ability to calculate the exact amount of losses data for future forecast. Plus, any task that have been assigned to particular team members does not limit the time to finish. These issues persist due to the lack of information management for managers to see the whole picture of the business process.

In recent years, business information system has been widely used in enhancing the quality of business processes. It can analyze the data, flow and the processes and help reduce the errors, costs. Besides, such data can be used to adjust the business plan and strategy. Most business information systems have been developed and considered with the concept of business process management [2,7]. Based on this perspective, this study analyzed the situation and problem of crafting process in emerald factories, and redesign the process based on business process management strategy in order to (1) reduce time cost, (2) know working progress, (3) make decision from defect information, (4) get insights to working performance, based on the concept of business process management. The information system was then proposed to help managing time, collecting data, and generating visual graphs for decision making.

The following sections of this paper are organized by firstly presenting the related concepts and studies in Section 2, followed by the method of redesigning the business process with the structure of information system in Section 3. The design and development of information system were then proposed where its evaluation was conducted and the result was revealed accordingly in Section 4. At the end of this paper, the discussion and conclusion of this study were described in details in Section 5.

2. RELATED STUDY

2.1 Business Process Management

Business Process is the processing of products or services from the organization or business which deploys resources for its customers that must be performed as a step by step. It also can be called as a set of steps or activities which will be the answer for organizations objective and policy. As the business process, the organization that produced the products and services for customers does not always need to be a business organization, it can be referred to any organization that has inputs and transforms those inputs into outputs [1]. Meanwhile, business process management is a principle of developing, operation, performance measurement. It is needed for enhancing the performance of the business processes continuously since starting to ending of the processes. Including processes development methodology and tools used for processes development. In addition, the business process management can be used to help the organization for reducing the outgoing that will cost through the processes and performance, also improves the quality and reduces mistakes that can cause in business processes.

There is a research from business process management to customer process management [6]. They are argued that in order to achieve customer centricity through business process management (BPM), companies have to obtain a profound understanding of customers' processes and when necessary change not only the interactions with but also the processes of their customers too. In the last example, the research of measuring Business Process Management (BPM) using business process orientation and process improvement initiatives [3].

2.2 Business Information System

Nowadays, information system plays an important role in business for many organizations all over the world. With information system supporting the business, it helps to organize the processes to perform as a system. It is designed and developed to operate the business plan as well as support all of the organizations to work along together with the data effectively in both operation and management level. With modern technologies in a present, business or organization with an ability to adapt itself with the information system, in order to handle the processes of products or services that will win over the customers and have the advantages over the competitor. Moreover, with a capability of classifying the business information system, it helps to improve the performance of the organization which helps the organization has the advantages in a competitive market and it also has the potential for helping the organization in business growth too.

In the business information system, there are different processes that are applied to the organization to support the operation of the business in a systematic manner, e.g., Transaction processing systems (TPS), Decision Support systems (DSS), and Management Information Systems (MIS). The difference between DSS and MIS is to focus on structured decisions and use internal data from the DSS system. The aim is to manage (supervise) the work of the operating unit. Executive Support System (ESS) is another type of decision support system developed specifically for people high-level management. The business information system responses with the business or organization without disturbing with the organization business workflow. It can be said that the business information system has the potential for solving, supporting, and improving the organization performance. Using an executive information system (EIS) to help decision support by the creation of information systems that enhance the managers'

knowledge of the environment where the organization operates [4,5,8].

3. METHOD

3.1 Business Process Understanding

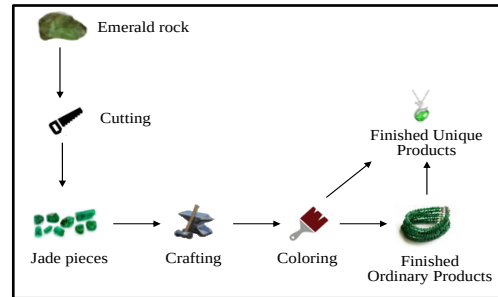


Figure 1. An illustration of major processes in an emerald factory.

In this type of business, most companies often find vacant lands which are believed to have many natural resources buried underground. As shown in Figure 1, the emeralds acquired are transported to the head office in town to start the crafting processes, making shapes and then deliver to the customers and the leading demand market. The company is responsible for mining emeralds, making them into valuable shapes, and bringing them into the market. The emeralds are sent to the crafting factory to begin the cutting and shaping processes. Each emerald-contained rock is analyzed for the usable emerald residing in the rock in the pre-crafting section and afterwards, the rock is cut into pieces to for the next crafting process. The managers will decide which parts of the rock will go to which crafting section. Each jade shape has its own unique processes in the crafting section. The finished products are then sent to coloring process and finally prepared for transportation.

3.2 Business Process Redesign

To better understand how the crafting process works, an as-is swimlane diagram, shown in Figure 3. Each rock sent to this factory is attached with a unique barcode, measured certain weights, determined parts to be cut, and then cut down into pieces. Each piece type also has its own barcode to remember the number of jades and to prevent losing. The managers will decide which section the product will go to depending on size and quality of jades, and the image designer will play a role if there are unique shape of jades are concerned to be crafted. Each part is sent to the specific crafting machine for specific shapes that has been determined previously. The finished products are then colored to make it look brighter and then packed for transportation. Being mentioned above, the managers do not get involved in the core processes and no specific times are given to employees to finish the job. There will also be few jades broken during the crafting process, but there is nothing to report the defect statistics quarterly or yearly. Therefore, these mentioned issues need to be redesigned to improve the crafting process.

In this study, A to-be swimlane diagram of crafting process has been proposed in Figure 4 to change the work flow and activity of the industrial processes. It is the improved swimlane from the previous one and include the information system to assist in the business process for managers to make the right decision, shown in the circle areas. Besides, the system helps setting times during which certain processes have to be done, where in the as-is processes, there is no time limited for any process to finish the job.

The proposed system helps in times limiting during which processes have to be done. It allows managers to set different deadlines to different types of products and managers can start or stop a process via phone to allow crafting employees know when to start and stop the process. This will make the work flow efficiently and reduce time cost. The information system can also gather the defect information input from software staff and generate it into reports for managers to make the right decision.

3.3 Business Information System Development

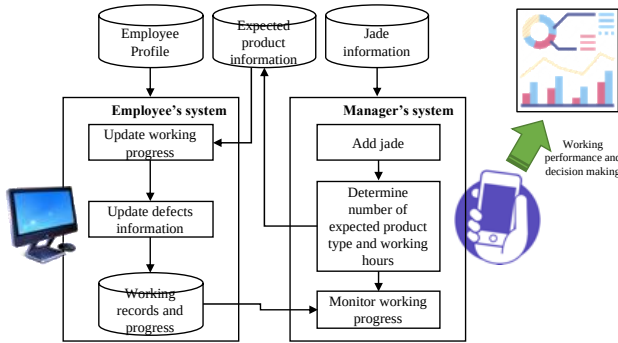


Figure 2. An overall structure of the proposed information system.

After the business process was redesigned, it is necessary to improve the process by employing the benefits of information system. As shown in Figure 2, there are two different roles: manager and employee. This system starts with the manager by adding jade into the system and determine the number of expected types and products, together with working hours assigned to the

employees. This information is kept into the database, where the employees can update the working progress and report the defects information found during the process. All information is kept in the database and displayed on the manager's system. This way, managers do not have to worry about late production whenever he is not around. The manager can employ the recorded transactions to further analyze the working performance of employees and making decision on other business aspects.

The proposed information system was developed as a progressive web-based application, where the employees can conveniently use on the desktop, and the manager can friendly use on the go with mobile device. It is deployed on the cloud hosting server run with Ubuntu 18.04 open-source Linux operating system with daily backup support.

4. RESULTS

4.1 System Result

In this section, some screenshots of the proposed system were presented. Figure 5 (left) shows how the manager can add the jades into the system via mobile by scanning the barcode of original jade rock. Then the manager can determine the expected type of product from this jade with assigned shapes and amount. Also, the expected hours for this process done by the employees. Once determined, the employees can update the progress of each assigned type by entering the finished number of product and defects found, see Figure 5 (top right). At the bottom of the page, the employees can see the remaining amount of this process.

The manager can see the status and progress of work assigned on the mobile, as shown in Figure 5 (bottom right). This will give the manager the overview of work performance in his factory and can be used for employees' performance analysis and improvement of

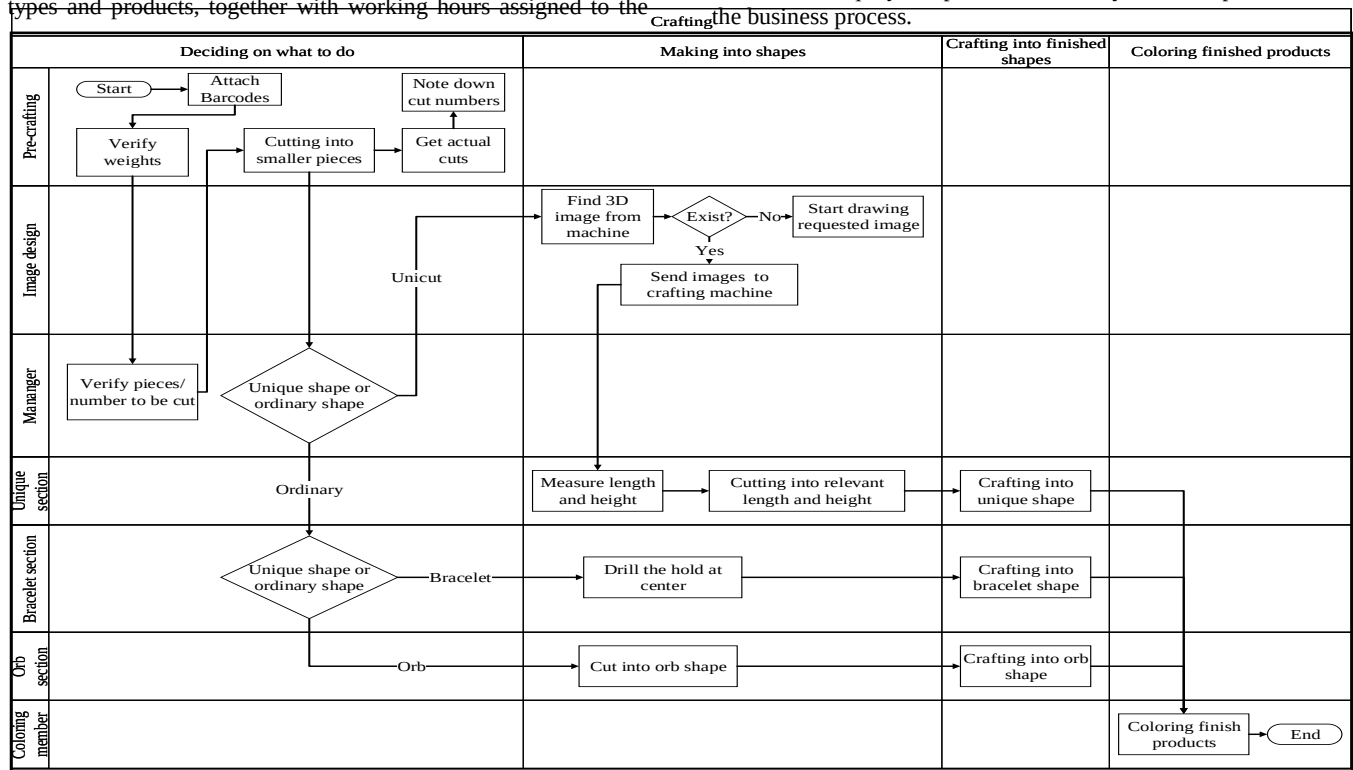


Figure 3. An as-is swimlane diagram of crafting process.

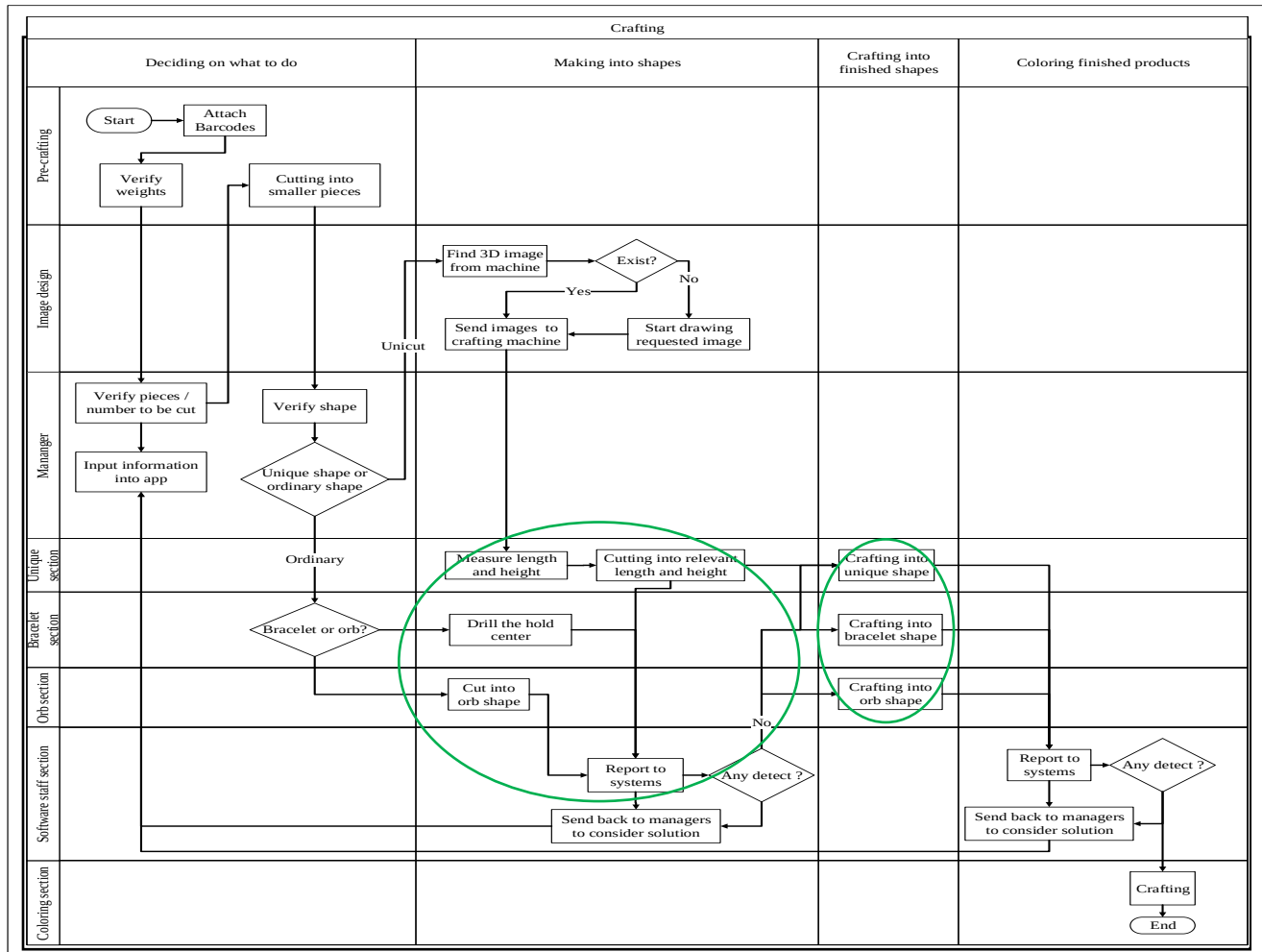


Figure 4. A to-be swimlane diagram of crafting process.

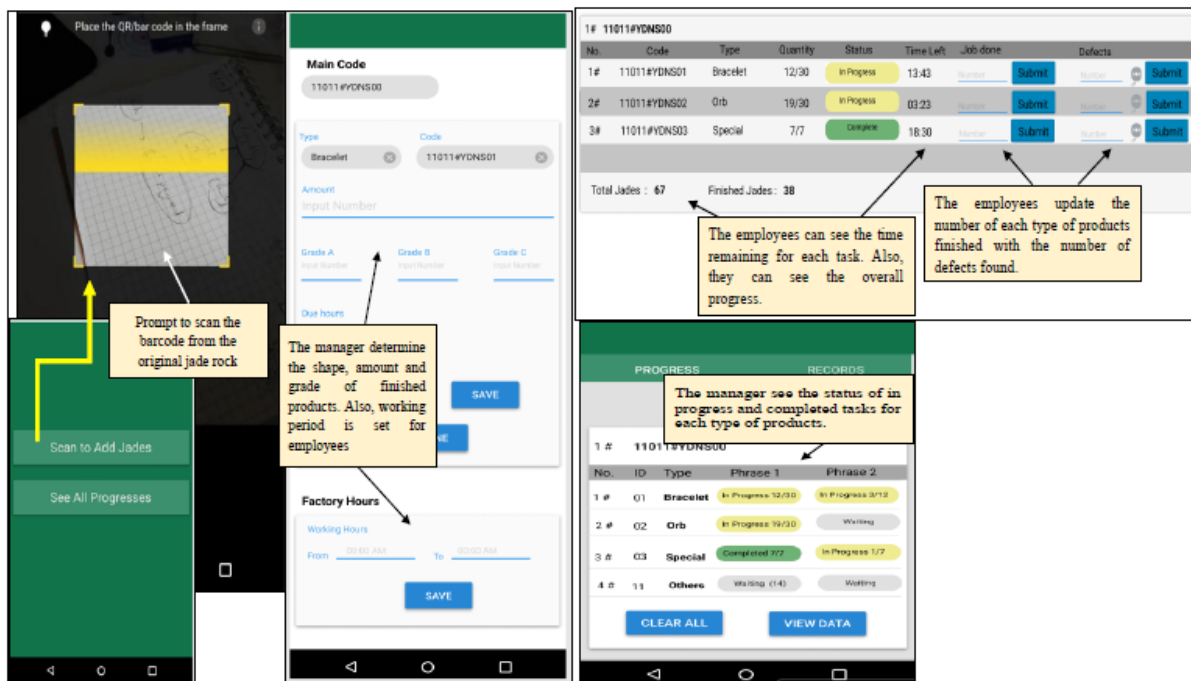


Figure 5. Sample screenshots of the proposed information system.

4.2 Acceptance of the Proposed System

After the proposed system has been used for weeks in the factory, the system evaluation was conducted on both employees and manager via the means of 5-point Likert scale questionnaire on the system acceptance and the attitudes towards using the system. Please note the only few workers were included in this study due to their time availability.

Table 1. Acceptance of the proposed system

Dimension	Employee s (n = 16)	Manager (n = 5)	Average
Ease of Use (EOU)	3.87	4.12	3.40
Usefulness (UNS)	4.39	4.58	4.49
Attitude towards Using (AUS)	4.33	4.46	4.40
Intention to Use (ITU)	4.21	4.52	4.37

As shown in Table 1, it was found that both employees and managers rated high degree of accepting this system ranging between 3.87 to 4.39 and 4.12 and 4.58, respectively. It was clear that the managers had higher average scores on all four dimensions. This can be considered that they felt the system provided them more business insights. However, the employees felt that how to use the system was not that convenient.

4.3 Attitudes towards the Proposed System

In addition to the acceptance of the proposed system, it is necessary to measure how the attitudes of the managers have been changed towards the ECP after experiencing the system. Based on the analysis results shown in Table 2, it was found that the managers had significantly better attitudes towards ECP on all aspects from time and cost saving, monitor working progress, working performance analysis, and defects analysis. This strongly confirms that the proposed system could enhance the possibility and effectiveness to their emerald business process.

Table 2. A comparison of managers' attitudes towards ECP with and without the proposed system

Aspect	Before (M ±SD)	After (M ±SD)	t
Time and cost saving	3.56 ±0.13	4.41 ±0.21	5.96*
Monitor working progress	2.78 ±0.31	4.72 ±0.15	9.75***
Working performance analysis	2.65 ±0.07	4.32 ±0.18	19.33***
Defects analysis	2.54 ±0.72	4.41 ±0.67	4.25*

* $p < 0.05$, *** $p < 0.001$

5. DISCUSSION AND CONCLUSION

5.1 Discussion

Owing to the emerald crafting process in emerald factory, there are several limitations and difficulties that need to be improved. In this study, the business process of ECP was analyzed and redesigned for more effectiveness, while the corresponding information system has been proposed for the managers and employees. The system enables the possibilities in reporting and monitor the working progress, while the crafting hours for each product types can be determined. Moreover, the employees can simply update the completed tasks and report the defects. Based on the evaluation results, both employees and managers highly

accepted the system on all dimensions. Moreover, the managers had higher attitudes towards ECP after using the system.

With the findings of this study, it really helps the managers to gain insights to the effectiveness of ECP based on the data generated during the work process. Besides, it helps the manager to investigate whether the ECP is cost- and time- effective, and the performance of employees in the factory. Importantly, the manager has possibilities to analyze the factory hours in completing each type of products, including the defects and leftovers. This data can be used to improve the process accordingly. Regarding the employees, they can adjust their crafting performance upon the assigned time; otherwise, the manager can reconsider the time assigned [9]. However, it is important for the emerald factories to provide a series of training to the different levels and background of the users. It is also required to customize some features and interface to meet the actual business context. In the meantime, it is suggested to cultivate all workers with data vision as they may have to be more open and understand the importance of business improvement with data-associated system.

5.2 Conclusion

Although the proposed system could enhance the effectiveness in crafting process, some limitations can be observed. First, the current system cannot capture or record the photo of the working process and defects. Second, the employees do not know their own performance. Lastly, it is difficult to determine the steps in using the system. Therefore, the above limitations need to be addressed further.

Finally, several future works are recommended to extend the benefits of the proposed system. It could be extended to cover other processes in the emerald business, e.g., customer relationship management, sales and marketing. Data analytics and visualization techniques can be useful in giving more business insights visually and adaptively. External sensors and machine learning can be employed to predict the factory hours, recommend the arrangement of logistics, and prioritize the tasks to be done.

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